

KB

CONTACT

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SKILLS

Python · PyTorch · JAX ·
Transformers · Hugging Face ·
Fairseq · Deepspeed · Megatron-LM
· CUDA · NCCL · MPI · Kubernetes
· Slurm · AWS SageMaker · GCP
Vertex AI · RLHF · PPO · DPO ·
LoRA · QLoRA · PEFT · Model
distillation · INT8 quantization ·
TensorRT · ONNX · spaCy · NLTK

LANGUAGES

English – Native
German – B2

CERTIFICATIONS

Co-author, 'Scaling Laws for RLHF
in Large Language Models', ACL
2023
Inventor, US Patent 11,845,678:
'Efficient LoRA Adaptation for
Transformers', 2022
First author, 'JAX-Pallas Mixed
Precision Training for LLMs',
NeurIPS 2022

Katie Brown

Principal NLP Researcher

PROFILE

Senior researcher with extensive experience leading large-scale language model projects across academia, big-tech research labs, and AI startups. Expertise spans transformer architecture design, distributed training on GPU/TPU clusters, RLHF alignment, and efficient fine-tuning techniques. Proven track record of delivering latency-optimized models that power products serving millions of users and generating multi-million-dollar revenue streams.

EXPERIENCE

Principal NLP Researcher, Meta AI

2024 – Present

London, United Kingdom

- Led a team of 8 researchers to deliver a 70B-parameter multilingual LLM that achieved 12% lower latency on A100 GPUs via kernel fusion and INT8 quantization, deployed in an internal assistant serving 2M daily active users.
- Designed a mixed-precision training pipeline using JAX and Pallas that cut training FLOPs by 18% while keeping perplexity within 0.2 of baseline on the Pile dataset.
- Authored an internal RLHF-with-AI-feedback study that increased win-rate against a GPT-4 baseline by 7% on human preference tests, shaping the 2025 product roadmap.
- Secured \$3.2M of internal funding for infrastructure, negotiated procurement of 256 H100 GPUs and optimized Slurm scheduling to reach 92% cluster utilization.

Senior Research Scientist, PolyAI

2020 – 2024

London, United Kingdom

- Built an end-to-end speech-to-intent pipeline using a Conformer encoder and a 1B-parameter LLM, reducing latency from 820 ms to 460 ms on V100 GPUs and enabling real-time handling of 150 concurrent calls per instance.
- Pioneered a LoRA-based fine-tuning method with only 0.3 % trainable parameters, matching full-fine-tuning accuracy while cutting GPU memory usage by 64 %.
- Mentored four junior researchers and interns, two of whom were promoted to research scientist roles within 12 months.
- Published three ACL-workshop papers on efficient inference, collectively cited 110 times by end of 2023.

Research Scientist, Google Research

2015 – 2020

London, United Kingdom

- Contributed to the development of the T5-XL model, implementing distributed training on TPU v3 pods that reduced training time from 14 days to 9 days for a 11B-parameter checkpoint.
- Designed a data-filtering pipeline that removed 22% of noisy web-text while preserving downstream task performance, saving ~1.4PB of storage.
- Led a cross-team effort to integrate sparsely-gated Mixture-of-Experts layers into the Transformer, achieving a 1.4× throughput increase at equal accuracy.

- Authored 5 conference papers (NeurIPS, ICML, EMNLP) with combined citations exceeding 850.

Postdoctoral Research Fellow, University of Edinburgh

2012 – 2015

Edinburgh, United Kingdom

- Developed a neural transition-based dependency parser that attained 94.2% UAS on the Penn Treebank, setting a new state-of-the-art for parsers without external embeddings.
- Released an open-source toolkit for multilingual word-embedding alignment, downloaded over 12,000 times in its first year.
- Supervised three MSc projects; all students graduated with distinction and two continued to PhD programs.
- Awarded a £150k EPSRC grant to investigate syntax-aware language modeling, resulting in a journal article in TACL.

EDUCATION

University of Edinburgh

PhD, Computational Linguistics · 2008–2012